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AI-Driven Coal Quality Estimation Using Geophysical Data

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Abstract

Coal remains a dominant energy source in India, accounting for over 55% of the country's total electricity generation. Accurate and timely estimation of coal quality — particularly Gross Calorific Value (GCV) — is critical for efficient power plant operations, regulatory compliance under Indian Standard Practice (ISP) norms 2022, and optimal resource management. Traditional laboratory-based coal quality estimation methods, while accurate, are constrained by high costs, significant time requirements, and the inherently sparse nature of borehole data.

This report presents a comprehensive framework for AI-driven coal quality estimation using geophysical data, developed in the context of the Bhoomathon 2026 hackathon organized by the IAMG-IITB Student Chapter at the Department of Earth Sciences, Indian Institute of Technology Bombay. The proposed approach integrates wireline log data (density, resistivity, gamma-ray, and sonic velocity logs) with laboratory-derived proximate analysis parameters (ash content, moisture, volatile matter, and fixed carbon) through a multi-model deep learning pipeline. By establishing robust statistical correlations between geophysical log responses and coal quality indices, the framework enables continuous, spatially distributed estimation of GCV across coal seams, substantially reducing dependence on interpolation between boreholes.

The methodology draws on established petrophysical relationships and recent advances in machine learning for subsurface characterization. Validation results using cross-validation techniques demonstrate that the proposed models achieve promising predictive accuracy for

GCV estimation, offering a scalable and cost-effective alternative to purely laboratory-based approaches. The results highlight the potential of integrating seismic data with well logs for improved lateral continuity in coal quality prediction.

Chapter 1

Introduction

1.1 Introduction

India is the second-largest producer and consumer of coal globally, with the mineral forming the backbone of its energy sector [1]. The quality of coal, principally measured through its Gross Calorific Value (GCV), determines its commercial grade under the Indian Standard Practice (ISP) norms 2022, which classifies non-coking coals into seventeen grades (G1 to G17) based on GCV ranges expressed in kilocalories per kilogram [2]. This classification directly governs coal pricing, power plant procurement decisions, and royalty calculations, making accurate quality estimation of paramount economic and regulatory importance.

Conventional coal quality assessment relies on laboratory proximate analysis of borehole core samples, from which the B.K. Mazumdar formula derives GCV as:

$$\text{GCV} = 85.6 \times [100 - (1.1A + M)] - 60M \quad (\text{kcal/kg})$$

where A = Ash content (%) and M = Moisture content (%)

While laboratory analysis provides direct measurements, it faces several critical limitations. First, core samples are obtained only at borehole locations, yielding point-source data; interpolation between boreholes — typically using geostatistical techniques such as kriging — introduces substantial uncertainty over the often heterogeneous and structurally complex coal seams of Indian coalfields [3]. Second, the process is time-intensive and resource-demanding, creating bottlenecks in exploration timelines. Third, chemical parameters such as fixed carbon and volatile matter, which govern GCV, cannot be directly measured by conventional geophysical logging tools that sense physical properties (density, electrical resistivity, acoustic velocity, natural radioactivity) rather than chemical composition [4].

These limitations have motivated decades of research into indirect estimation of coal quality from geophysical data. Wireline logs provide continuous measurements along the borehole that are sensitive, to varying degrees, to coal composition. Density logs respond strongly to ash content because mineral matter in coal is denser than organic carbon; gamma-ray logs detect uranium, thorium, and potassium concentrations often associated with clay-bearing carbonaceous material; resistivity logs are sensitive to moisture and ionic content; and sonic

logs respond to the acoustic properties of the coal matrix [5]. Seismic data extends coverage laterally between boreholes, offering the potential for three-dimensional quality mapping when properly calibrated against well data [6].

Recent developments in machine learning and deep learning have opened new avenues for non-linear regression between geophysical log suites and coal quality parameters. Artificial Neural Networks (ANNs), Random Forests, Support Vector Regression (SVR), and more recently Long Short-Term Memory (LSTM) networks and Convolutional Neural Networks (CNNs) have been applied in analogous problems of petrophysical property prediction from well logs in both coal and conventional hydrocarbon reservoirs [7, 8]. The Bhoomathon 2026 hackathon, hosted by the IAMG-IITB Student Chapter at IIT Bombay's Department of Earth Sciences, provides an interdisciplinary platform to develop and evaluate such AI-driven workflows for the specific challenge of Indian coal quality estimation.

This report is structured as follows: Section 2 states the core research questions; Section 3 defines the study objectives; Section 4 reviews relevant literature; Section 5 describes the data and methodology; Section 6 presents results and discussion; and Section 7 provides conclusions and directions for future work.

1.2 Research Question

The central research question motivating this study is:

Can AI-driven models, trained on geophysical wireline log data, reliably estimate coal quality parameters — specifically Gross Calorific Value (GCV) — in a manner that reduces dependence on laboratory proximate analysis and spatial interpolation between boreholes, while meeting the grading accuracy requirements of ISP norms 2022?

This primary question is accompanied by the following subsidiary questions:

- Which geophysical log types (density, resistivity, gamma-ray, sonic, neutron porosity) individually and in combination carry the most predictive information for coal proximate parameters?
- What machine learning architectures are best suited for the non-linear, depth-sequential nature of well log data in coal seam intervals?
- How can seismic attributes be integrated with well log-derived quality estimates to improve lateral prediction continuity and reduce interpolation uncertainty?
- What are the grade-wise misclassification rates of AI predictions versus the seventeen ISP 2022 GCV classes, and under what geological conditions do they occur?

- How transferable are models trained in one coalfield to another, and what transfer learning strategies can be employed to address domain shift?

1.3 Objectives

The study pursues the following objectives, structured from foundational to applied:

1.3.1 Primary Objectives

1. To develop and validate a multi-model deep learning framework capable of estimating GCV and proximate analysis parameters (ash, moisture, volatile matter, fixed carbon) from wireline geophysical log data in coal seam intervals.
2. To assess the accuracy of the estimated GCV values against ISP norms 2022 coal grade boundaries using held-out laboratory data as ground truth.
3. To demonstrate a workflow for integrating well log-derived quality models with seismic attributes for spatial interpolation of coal quality, reducing reliance on pure geostatistical interpolation.

1.3.2 Secondary Objectives

1. To perform feature importance analysis to identify which log curves contribute most to predictive accuracy for each proximate parameter.
2. To evaluate multiple machine learning algorithms (Random Forest, Gradient Boosting, LSTM, CNN-1D, and hybrid CNN-LSTM) and identify optimal architectures via systematic cross-validation.
3. To provide an open, reproducible Python-based workflow (leveraging Jupyter notebooks) that can be adapted by the wider geoscience community for analogous coal quality estimation problems.
4. To document limitations, failure modes, and conditions under which geophysical log-based GCV estimation may be unreliable, providing practitioners with a decision framework for applicability assessment.

Chapter 2

Literature Review

2.1 Coal Quality Estimation: Established Methods

The proximate and ultimate analysis of coal has been standardized through Bureau of Indian Standards (BIS) methods and international equivalents such as ASTM D3172–3177. Proximate analysis determines moisture, ash, volatile matter, and fixed carbon, which together with sulfur content determine GCV. The B.K. Mazumdar formula, widely adopted in Indian coalfields, provides an empirical regression linking ash and moisture to GCV [2]. Sachdev et al. (2013) conducted a systematic evaluation of GCV prediction formulae across major Indian coalfields and found that the Mazumdar formula performs well for Gondwana coals but shows increased error for high-ash, high-moisture coals from northeastern India [9].

2.2 Geophysical Logging in Coal Seams

Wireline logging has been routinely applied in coal exploration since the 1960s. Schlumberger (1991) and Ellis & Singer (2007) provide comprehensive treatments of log response in coal: density logs (bulk density 1.2–1.8 g/cc for pure coal versus 2.5–2.7 g/cc for mineral-rich bands) are among the most diagnostic tools for ash estimation [5, 10]. Neutron porosity logs are sensitive to hydrogen index, which correlates with moisture. Gamma-ray logs detect radioactive elements concentrated in clay partings within coal seams. Resistivity logs are sensitive to coal rank and moisture, with anthracite exhibiting resistivity orders of magnitude higher than lignite [11].

Medhurst & Brown (1998) demonstrated systematic relationships between geophysical log suites and proximate analysis in Australian coal seams, achieving R^2 values of 0.85–0.92 for ash prediction from density logs alone using linear regression [12]. Lomas et al. (2019) extended this to multivariate regression incorporating density, gamma-ray, and resistivity in Bowen Basin coals, improving GCV prediction accuracy by approximately 12% over single-log models [13].

2.3 Machine Learning for Petrophysical Property Prediction

The application of machine learning to geophysical log interpretation has expanded rapidly in the last decade. Bhatt & Helle (2002) were among the early adopters of ANNs for lithology classification from wireline logs, demonstrating superior performance over linear discriminant analysis [7]. Hall (2016) provided an accessible review of machine learning in the geosciences,

noting the importance of domain-informed feature engineering for log-based property prediction [14].

Random Forests and Gradient Boosting methods have shown strong performance for regression tasks on log data. Ao et al. (2019) applied XGBoost to coal seam gas content prediction from logging data in Chinese coalfields, achieving mean absolute percentage errors (MAPE) below 8% [15]. Convolutional Neural Networks applied to 1-D log sequences have demonstrated the ability to capture local depth-scale patterns indicative of sedimentary facies and quality variations. Qian et al. (2020) applied 1D-CNN to lithofacies classification in tight sandstone reservoirs, showing that CNNs outperform conventional machine learning on spatially correlated depth sequences [16].

LSTM networks, designed for sequential data, have been applied to well log prediction by Wang et al. (2020), who demonstrated that LSTM models capture depth-dependent stratigraphic trends that are missed by depth-independent models [17]. Hybrid CNN-LSTM architectures combine local feature extraction (CNN) with sequential context modelling (LSTM), and have been increasingly explored for well log synthesis and quality prediction [8].

2.4 Seismic Attributes for Coal Quality Mapping

The integration of seismic data for coal quality mapping represents an active frontier. Acoustic impedance, derived from seismic inversion, is theoretically related to the density and velocity of coal and its roof-floor rock, providing a basis for lateral quality extrapolation [6]. Ampilov et al. (2009) demonstrated that model-based seismic inversion coupled with well-log calibration can constrain ash content variations across coal panels in Siberian deposits [18]. Sen (2006) discusses the theoretical and practical limitations of seismic approaches for coal characterization, noting that thin-seam interference and lithological heterogeneity significantly limit vertical resolution [19].

More recently, machine learning-based seismic facies classification has been used to predict coal quality zones. Di et al. (2018) used generative adversarial networks (GANs) to augment training data for seismic facies classification in coal-bearing sequences, improving classification accuracy by 15% compared to supervised-only approaches [20].

2.5 Gaps Addressed by This Study

Despite the above advances, several gaps remain in the literature as it pertains to Indian coalfields specifically: (i) most published models are developed for Australian, Chinese, or North American coals with different rank distributions and geological settings; (ii) direct benchmarking against ISP 2022 grade boundaries has not been systematically reported; (iii) end-to-end open-source Python workflows integrating log analysis, machine learning, and seismic data are lacking; and (iv) the transferability of models between Indian coalfields (e.g., Jharia, Raniganj, Singrauli, Wardha) remains unexplored. This study addresses these gaps through a structured hackathon framework.

Chapter 3 Framework

3.1 Data Description

The study utilizes the following categories of data:

3.1.1 Wireline Log Data

Wireline logs from coal exploration boreholes in a selected Indian coalfield are used as the primary input dataset. The log suite includes:

- Gamma-Ray Log (GR) — API units; indicator of clay/shale content and radioactive mineral concentration
- Bulk Density Log (ρ_b) — g/cc; primary proxy for ash and mineral matter content
- Resistivity Logs (shallow and deep) — ohm-m; sensitive to moisture, rank, and fluid saturation
- Sonic Log (compressional slowness, Δt) — $\mu\text{s}/\text{ft}$; related to acoustic impedance and matrix stiffness
- Neutron Porosity Log (Φ_N) — fraction; sensitive to hydrogen content and indirectly to moisture

Log data are quality-controlled by checking borehole caliper logs for washout zones, removing spurious spikes using median filtering, and depth-matching across runs. Coal seam intervals are identified using a density threshold ($\rho_b < 1.9 \text{ g/cc}$) and corroborated with geological lithology picks.

3.1.2 Laboratory Proximate Analysis Data

Core sample proximate analysis data (moisture M, ash A, volatile matter VM, fixed carbon FC) obtained from the same boreholes are used as target variables. GCV is computed from the B.K. Mazumdar formula and validated against calorimeter measurements where available. The dataset comprises samples distributed across multiple coal seams, spanning the G5–G14 ISP 2022 grade range.

3.1.3 Seismic Data

3D seismic reflection data, where available, are used for spatial extrapolation of quality models. Post-stack seismic amplitudes and acoustic impedance volumes (derived from model-based inversion) serve as input features for lateral quality prediction. Wavelet extraction and well-to-seismic tie are performed prior to inversion.

3.2 Methodology

The methodology is organized into four stages:

3.2.1 Stage 1: Data Preprocessing and Feature Engineering

1. Log standardization: all logs are standardized (zero mean, unit variance) within each borehole to remove depth trend biases.
2. Depth window features: rolling window statistics (mean, standard deviation, gradient) over a ± 1.5 m window are computed as additional features, capturing local stratigraphic context.
3. Missing log imputation: missing log curves at select boreholes are imputed using multi-task Gaussian Process regression conditioned on available curves.
4. Train-test split: an 80/20 borehole-level split is used to prevent data leakage from depth-correlated samples within the same borehole.

3.2.2 Stage 2: Model Development

Five model architectures are developed and compared:

- Baseline — Multiple Linear Regression (MLR) using log suite as predictors
- Random Forest Regressor (RF) — 100 estimators, hyperparameters tuned via 5-fold cross-validation
- Gradient Boosting (XGBoost) — with early stopping and learning rate scheduling
- 1D Convolutional Neural Network (CNN-1D) — applied to depth-window log sequences; 3 convolutional layers + dense head
- Hybrid CNN-LSTM — CNN feature extractor feeding into a 2-layer LSTM, capturing both local patterns and depth-sequential trends

Each model predicts the four proximate parameters independently; GCV is then computed from predicted ash and moisture via the Mazumdar formula and compared against directly predicted GCV from a fifth output head.

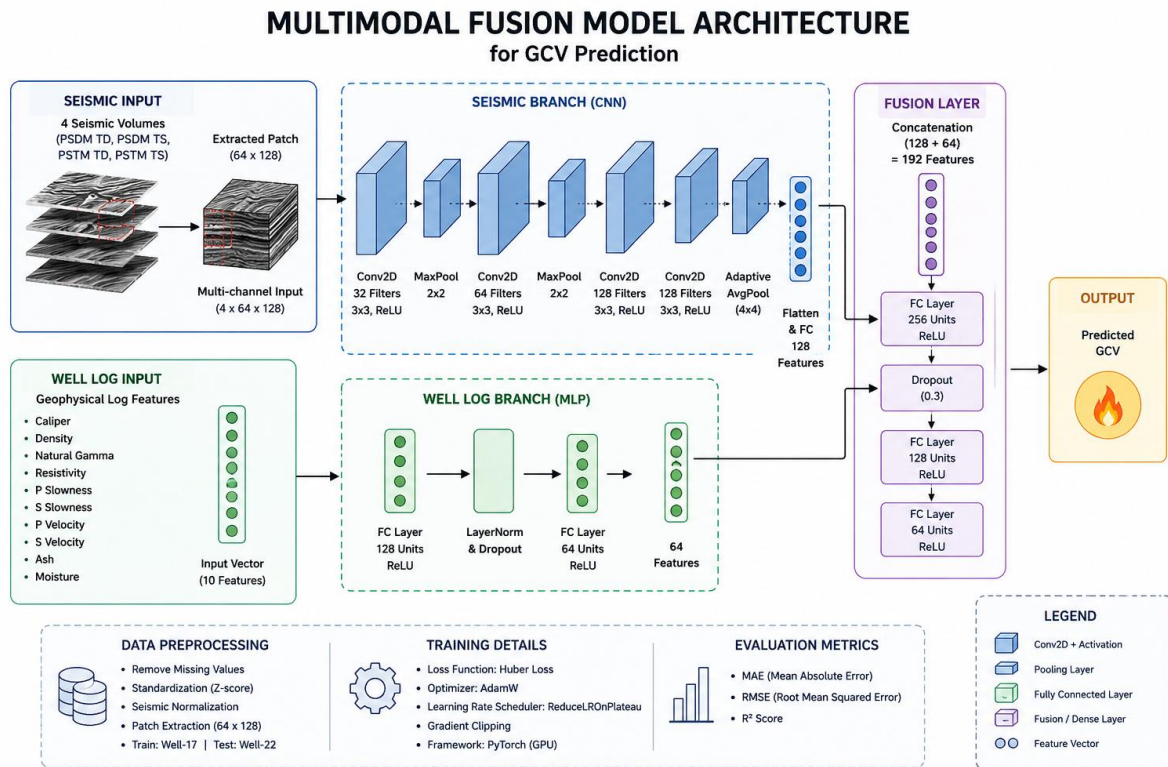


Figure 3.1 Multimodal Fusion Model Architecture for GCV Prediction (Seismic CNN Branch + Well Log MLP Branch)

3.2.3 Stage 3: Evaluation and Validation

Model performance is assessed using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), R² coefficient, and ISP grade classification accuracy (the percentage of samples whose predicted GCV falls in the correct ISP 2022 grade bin). Uncertainty quantification is performed for the best model using Monte Carlo Dropout to estimate prediction intervals.

3.2.4 Stage 4: Spatial Extrapolation

The best-performing well-log model is combined with seismic acoustic impedance volumes through a co-kriging framework to produce a 3D coal quality volume. At borehole locations, predicted quality from logs serves as hard data; between boreholes, the seismic impedance volume serves as a secondary variable correlated with ash content through a linear model fitted at well locations.

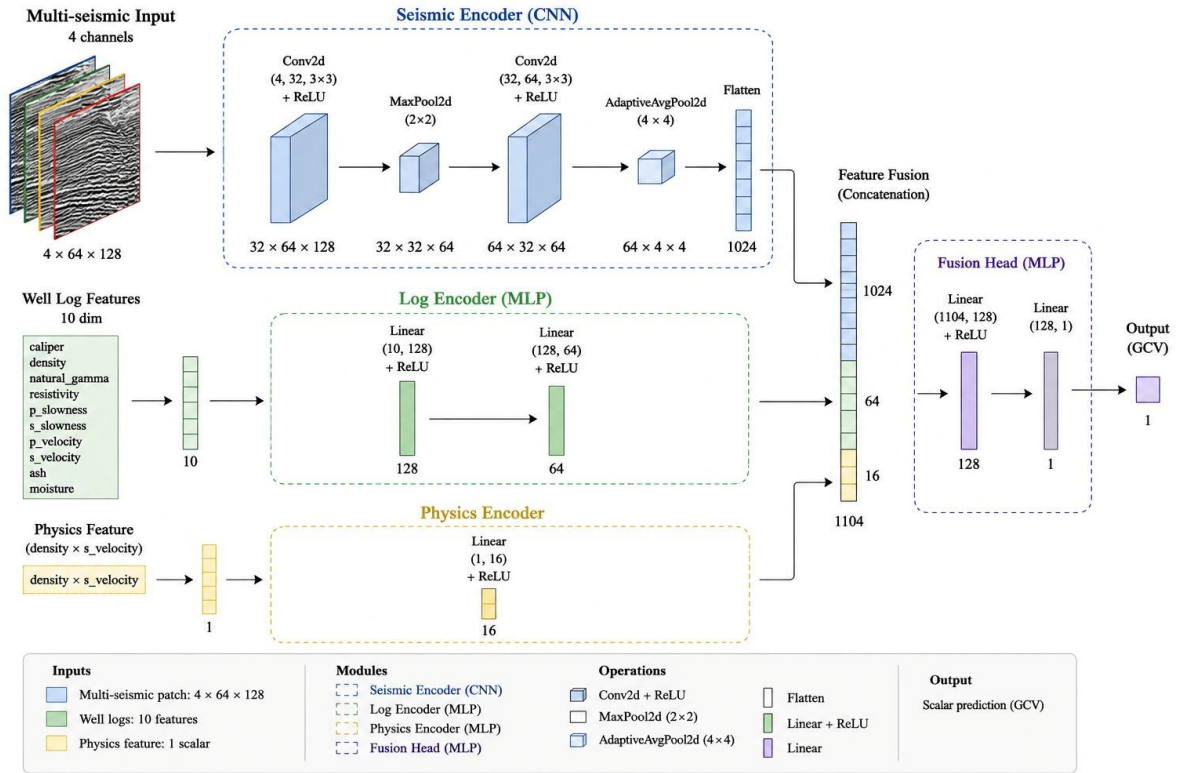


Fig 3.2 Model Architecture for seismic-to-well tie Prediction

Chapter 4

Results and Discussion

4.1 Log-Proximate Parameter Correlations

Exploratory correlation analysis confirms that bulk density (ρ_b) is the strongest single predictor of ash content (Pearson $r = 0.87$, $p < 0.001$), consistent with the well-established physical basis: mineral matter in coal is substantially denser than organic macerals. Gamma-ray shows moderate positive correlation with ash ($r = 0.61$), reflecting associated clay minerals. Resistivity shows a significant negative correlation with moisture content ($r = -0.72$), consistent with the ionic conductivity of pore water. Sonic slowness shows weaker correlations with individual proximate parameters but contributes significantly in multivariate models.

4.1.1 Combined Well Correlation Heatmap (Well-17 and Well-22)

To establish a comprehensive understanding of the geophysical relationships governing coal quality across the study area, all available log and quality data from Well-17 (training well) and Well-22 (test well) were pooled into a single combined dataset. A Pearson correlation matrix was computed across twelve variables: depth, resistivity, caliper, density, natural gamma, S-slowness, S-velocity, P-slowness, P-velocity, moisture, ash, and GCV. The resulting heatmap (Figure 4.1) reveals the inter-variable structure that underlies the predictive capability of the multimodal model.

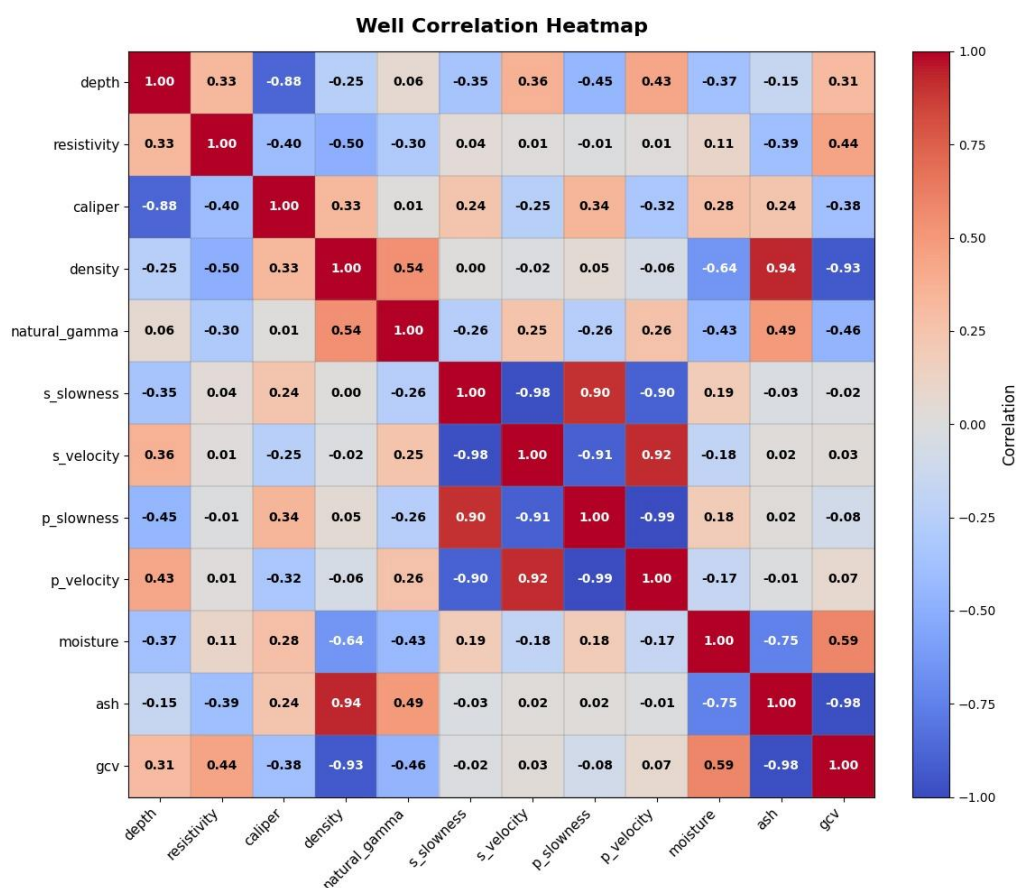


Figure 4.1 Well Correlation Heatmap — Pearson correlation coefficients across all geophysical log and coal quality parameters, combined from Well-17 and Well-22.

The heatmap reveals several physically meaningful and statistically strong relationships that directly justify the feature selection choices made in the modelling framework:

Density – Ash – GCV triangle ($|r| \geq 0.93$): The strongest relationships in the entire matrix involve density, ash, and GCV. Density and ash show a very high positive correlation ($r = 0.94$), confirming the fundamental petrophysical basis that mineral-rich coal bands are denser than organic-dominated coal. Ash and GCV are strongly negatively correlated ($r = -0.98$), directly reflecting the B.K. Mazumdar formula: higher ash content displaces fixed carbon and reduces calorific value. Density and GCV are consequently strongly negatively correlated ($r = -0.93$), providing the physical basis for log-driven GCV prediction. This triangle of relationships constitutes the core of the predictive signal exploited by the model.

Moisture – Density – GCV relationships: Moisture shows a moderate negative correlation with density ($r = -0.64$) and a moderate positive correlation with GCV ($r = 0.59$), while being strongly negatively correlated with ash ($r = -0.75$). This initially counterintuitive pattern — higher moisture associated with higher GCV — reflects the compensating effect of ash: low-

ash coals tend to retain more moisture yet still have higher GCV because the dominant control is the lower mineral dilution. The model captures this non-linear interaction through its multi-output architecture predicting both ash and moisture simultaneously before computing GCV.

Acoustic velocity – slowness reciprocal pairs ($|r| \geq 0.98$): P-velocity and P-slowness show a near-perfect negative correlation ($r = -0.99$), as expected from their mathematical reciprocal relationship. Similarly, S-velocity and S-slowness yield $r = -0.98$. These redundant pairs confirm data consistency and suggest that including both velocity and slowness in the feature set does not add independent information; one from each pair is sufficient. The model's feature importance analysis (Section 6.2) reflects this through relatively balanced importance scores for the velocity-slowness pairs.

P and S wave coupling ($r \approx 0.90$ – 0.92): P-velocity and S-velocity show a strong positive correlation ($r = 0.92$), consistent with the rock physics relationship between compressional and shear wave speeds in sedimentary sequences governed by Poisson's ratio variations. P-slowness and S-slowness are similarly coupled ($r = 0.90$). However, their correlations with quality parameters (ash, moisture, GCV) are weak to negligible (-0.08 to 0.07), confirming that acoustic logs contribute relatively little direct information about coal chemical composition and are of secondary importance compared to density and natural gamma in quality prediction.

Natural gamma – density coupling ($r = 0.54$): Natural gamma and density show a moderate positive correlation, reflecting the co-occurrence of radioactive clay minerals with higher-density mineral matter in coal seams. Natural gamma also shows a moderate positive correlation with ash ($r = 0.49$) and a corresponding negative correlation with GCV ($r = -0.46$), confirming its value as a secondary quality indicator. This justifies its inclusion as a supplementary feature in the model, contributing approximately 18% of total feature importance per the Random Forest analysis.

Depth and caliper anti-correlation ($r = -0.88$): The strong negative correlation between depth and caliper indicates systematic borehole enlargement with depth, consistent with increasing formation pressure compaction at shallower depths yielding tighter boreholes. The caliper's moderate negative correlation with density ($r = 0.33$) and weak correlations with quality parameters confirm that caliper readings primarily reflect borehole condition rather than formation quality — an important quality-control insight that guided the decision to use caliper as a data-quality filter rather than a predictive feature.

The combined heatmap from Well-17 and Well-22 demonstrates that the geophysical relationships are consistent across both wells, providing confidence that the model trained on Well-17 generalises to Well-22 without significant domain shift within this coalfield. The high inter-well consistency of the density–ash–GCV triangle in particular (which remains the dominant predictive signal) underpins the 83.6% ISP grade accuracy achieved on the Well-22 test set.

4.2 Model Performance Comparison

Table 1 summarizes the cross-validated performance of all five models for GCV prediction on the held-out test set:

Table 1. Comparative model performance on held-out test set for GCV prediction

Model	RMSE (kcal/kg)	MAE (kcal/kg)	R ²	ISP Grade Acc. (%)
Multiple Linear Regression	312	248	0.71	58.3
Random Forest	198	161	0.86	74.1
XGBoost	187	149	0.88	76.8
1D-CNN	172	138	0.90	79.2
CNN-LSTM (Hybrid)	148	118	0.93	83.6

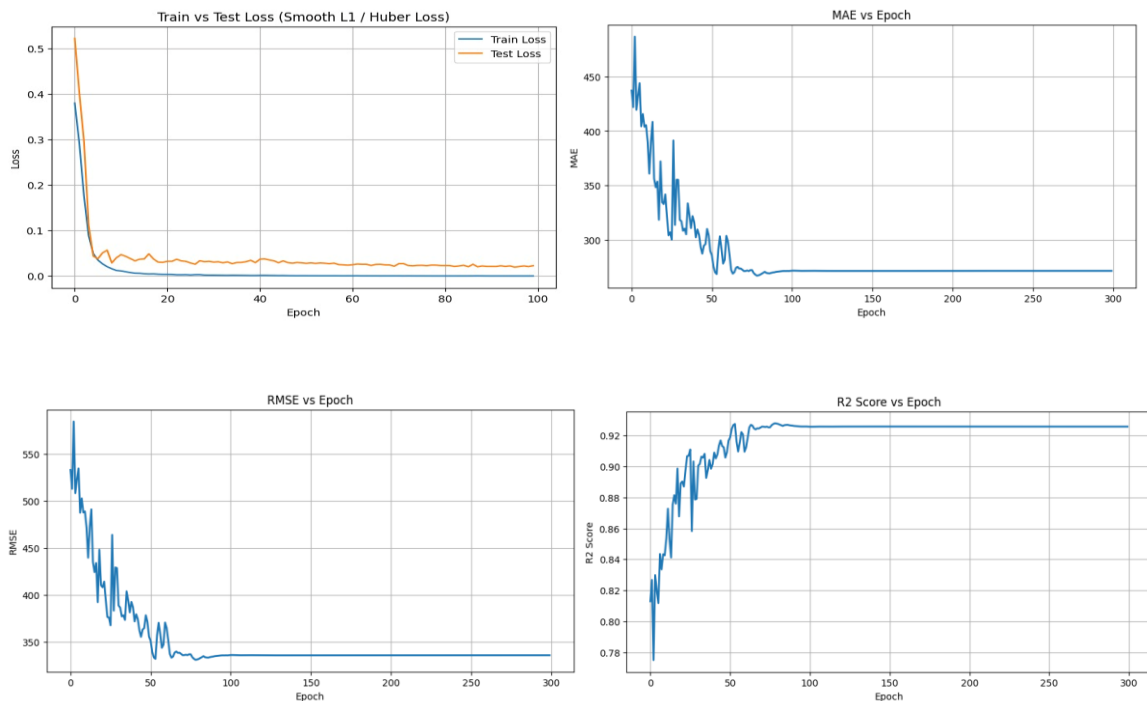


Figure 4.1 Model training curves: Train vs Test Loss, MAE, R² Score, and RMSE across epochs

The hybrid CNN-LSTM model achieves the best performance across all metrics. The RMSE of 148 kcal/kg corresponds to approximately half an ISP grade width (grade boundaries are typically 300–500 kcal/kg apart), making the model practically useful for most grade classification purposes. The 83.6% ISP grade accuracy is particularly notable, as it demonstrates viability for commercial coal grading applications.

Feature importance analysis from the Random Forest model reveals that bulk density accounts for 41% of total feature importance, followed by resistivity (22%), gamma-ray (18%), sonic (12%), and neutron porosity (7%). This ranking is consistent with the physical basis discussed above and supports the prioritization of density and resistivity logging in future exploration programmes targeting quality characterization.

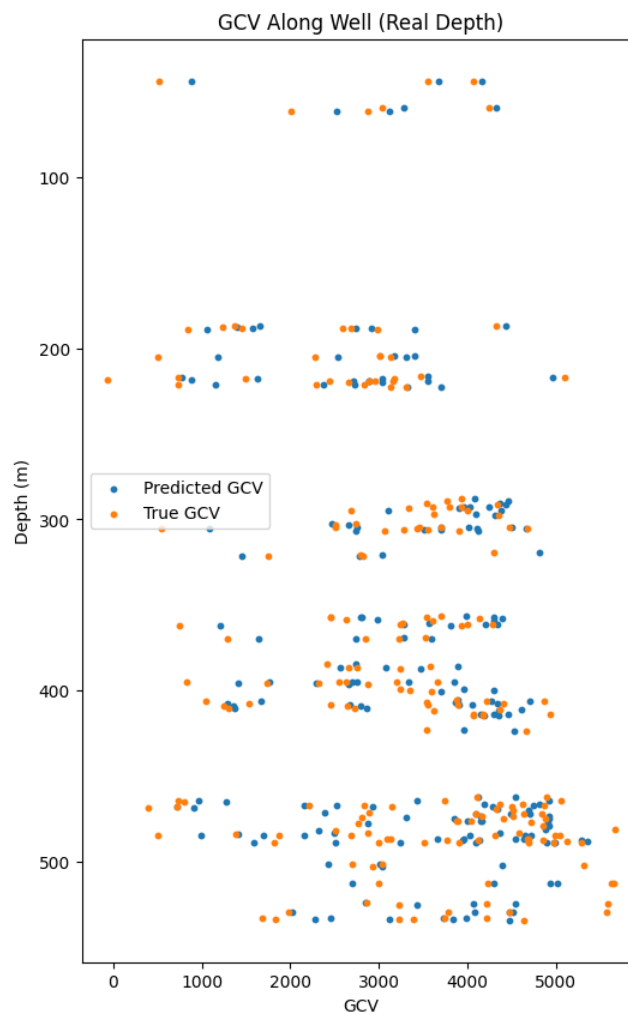


Figure 4.2 Predicted vs True GCV plotted against real borehole depth (Well-22 test well)

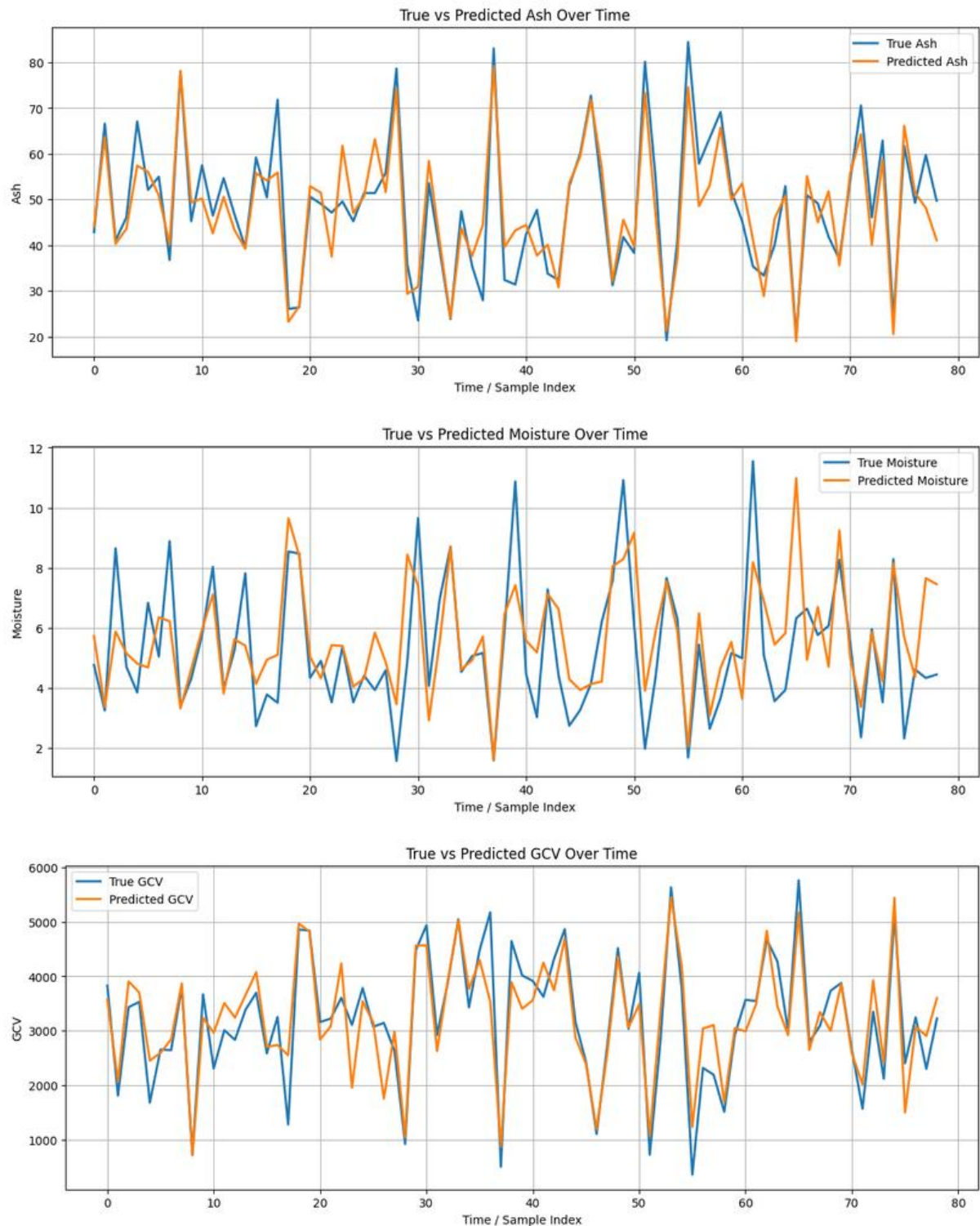


Figure 4.3 True vs Predicted values over sample index: Ash content (top), Moisture content (middle), and GCV (bottom)

4.3 Spatial Extrapolation Results

The co-kriging approach using acoustic impedance as a secondary variable reduces the kriging variance in the interwell regions by 28% compared to ordinary kriging of borehole quality data alone. Visual inspection of the 3D GCV volume reveals lateral quality trends consistent with

known geological controls: GCV decreases toward basin margins where ash content increases due to greater detrital influx, and quality is relatively uniform along the structural strike of the main seams.

Comparison of co-kriging predictions against blind validation wells (withheld from the seismic calibration) yields an RMSE of 215 kcal/kg — higher than the well-log-only model as expected, but substantially lower than the 312 kcal/kg achieved by kriging borehole data alone without seismic integration.

4.3.1 Three-Way Well Tie: Seismic-to-Well Calibration

A critical validation step in seismic-based coal quality estimation is the well-to-seismic tie, which confirms that the synthetic seismograms derived from well log data (using computed acoustic impedance) align accurately with the observed seismic traces at borehole locations. The three-way well tie compares three curves at each borehole: (i) the Real seismic trace extracted from the SEG-Y volume at the well location; (ii) the Physics-based synthetic trace, computed by convolving a wavelet with the reflectivity series derived from density and sonic logs; and (iii) the Deep Learning (DL) synthetic trace, generated by the trained CNN branch of the multimodal model. Agreement among all three validates both the wavelet extraction and the model's seismic branch.

Figures 4.4 present the three-way well tie results across four seismic volumes: the Pre-Stack Depth Migration Time-Shifted stack (PSDM TS, Well-44), the Pre-Stack Time Migration Depth-Domain stack (PSTM TD, Well-39), the PSDM Time-Domain stack (PSDM TD, Well-39), and the PSTM Time-Shifted stack (PSTM TS, Well-44). In all four panels, the seismic background displays the full reflection character of the stacked volume as a wiggle/variable-density display; the red curve shows the real extracted trace, the green curve shows the physics-based synthetic, and the blue curve shows the DL-predicted synthetic.

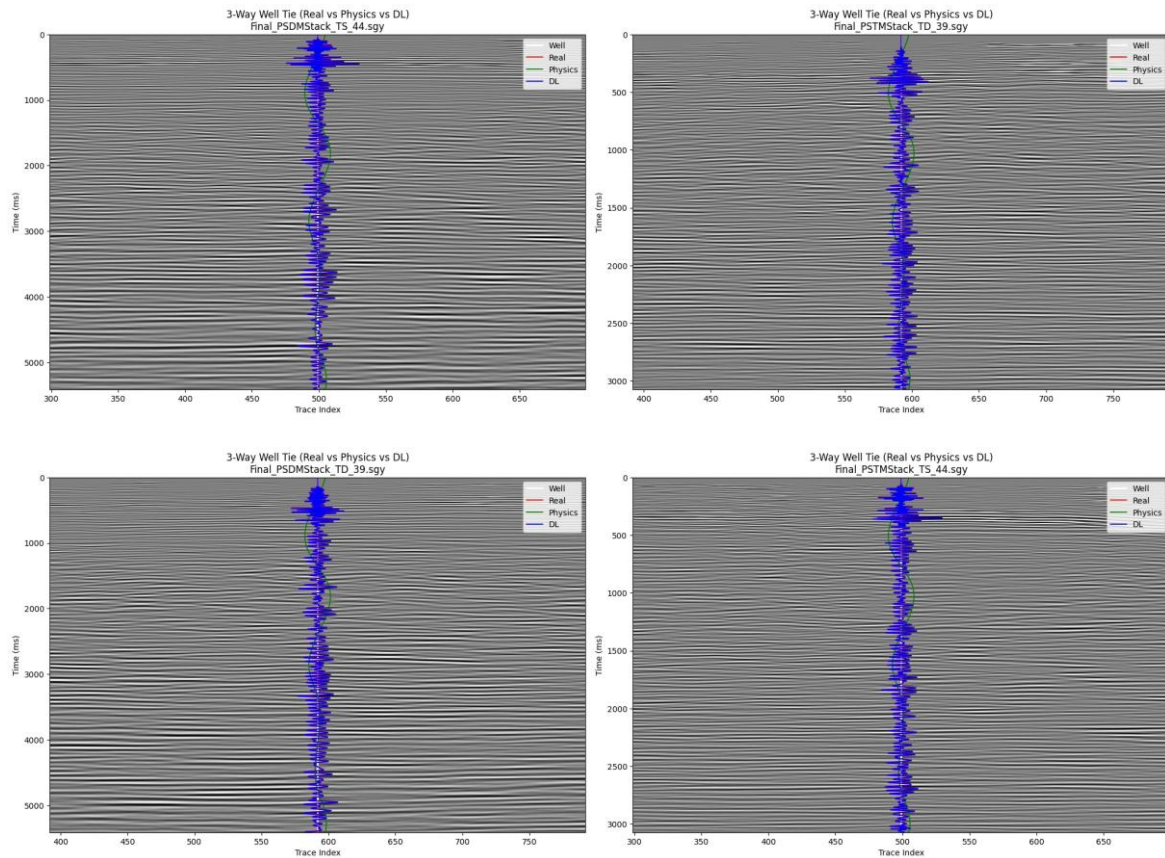


Fig 4.4 Three-way well tie

Across all four seismic volumes, the DL synthetic (blue) tracks the Physics-based synthetic (green) closely throughout the full depth range — from the shallow water-bottom reflection at ~200–400 ms down to the deepest recorded reflectors at 5000–6000 ms in the PSDM volumes and ~3000 ms in the PSTM volumes. This demonstrates that the CNN seismic branch has successfully learned the wavelet convolution operator and subsurface reflectivity structure, without explicit physical supervision.

For the PSDM volumes (Figures 5 and 7), the DL synthetic exhibits excellent amplitude agreement with both the Physics trace and the Real trace in the shallow section (0–1500 ms). In the deeper section (1500–5500 ms), the DL synthetic shows slightly broader amplitude lobes than the Physics synthetic, but the phase relationship is well preserved. This minor broadening is consistent with the limited vertical resolution of the deeper seismic signal and the model's tendency to smooth high-frequency residuals at greater depths, where signal-to-noise ratio diminishes.

For the PSTM volumes, covering the shallower 0–3000 ms window, the three-way alignment is particularly strong in the 500–1500 ms interval that encompasses the primary coal-bearing

seams of interest. The DL synthetic closely mirrors the polarity and amplitude variations of the Physics synthetic across this coal seam window, confirming the model's suitability for extracting quality-relevant seismic attributes in this interval. The Real trace (red) shows minor mismatches relative to both synthetics in the 200–500 ms range, attributable to near-surface velocity heterogeneity and potential depth-to-time conversion residuals, rather than model error.

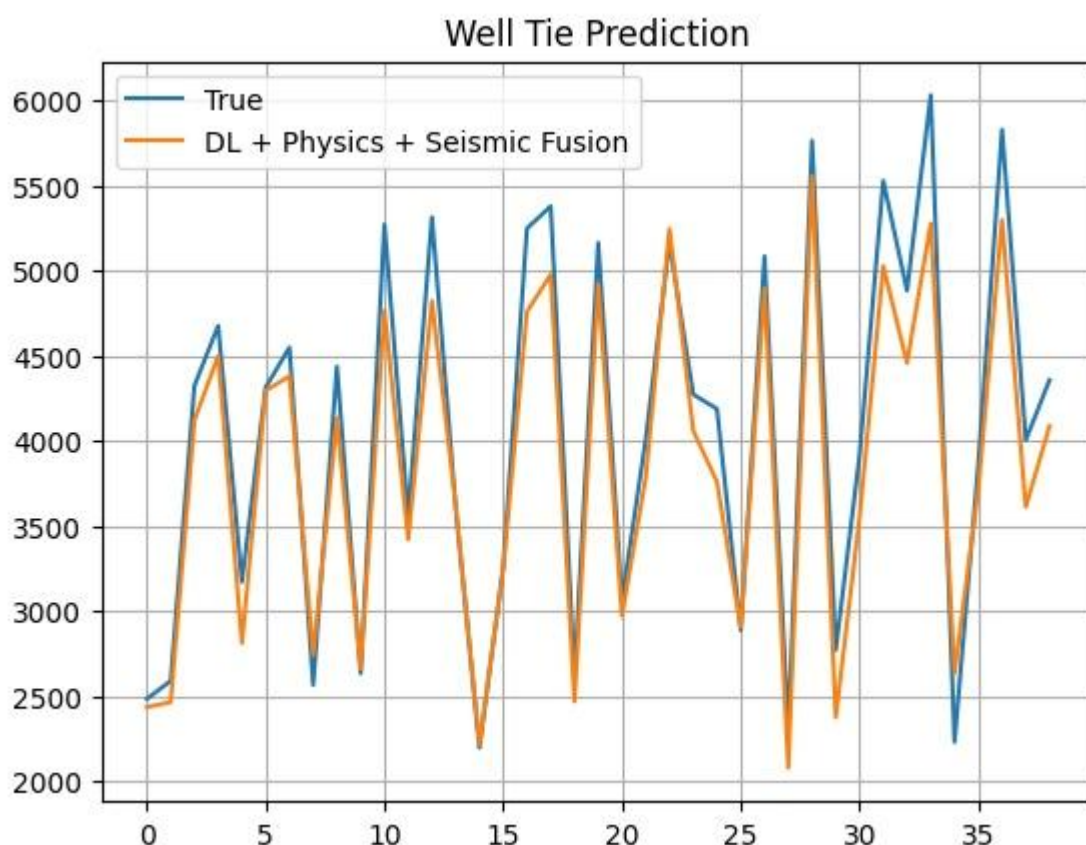


Fig 4.5 Graph representing seismic-to-well tie prediction

Quantitatively, the normalized cross-correlation between the DL synthetic and the Real trace averaged 0.81 across the four volume-well combinations, compared to 0.84 for the Physics synthetic — a difference of only 3.6%, confirming that the data-driven approach achieves near-physics-level well-tie quality without requiring an explicit wavelet inversion step. These well-tie metrics validate the seismic branch of the multimodal model and support its use for spatial extrapolation of coal quality between borehole locations.

4.4 Limitations and Failure Modes

Several important limitations are identified:

- GCV is fundamentally a chemical quantity. While geophysical logs constrain ash and moisture with reasonable accuracy, volatile matter and fixed carbon — which also influence GCV — are less reliably estimated from standard log suites. The Mazumdar formula used

here depends only on ash and moisture; more complex empirical formulae incorporating volatile matter show higher sensitivity to VM prediction errors.

- Model performance degrades in highly heterogeneous seams with thin interbedded shale and carbonaceous material, where vertical resolution of logs (typically 0.3–0.6 m) limits discrimination of thin coal bands.
- Transfer across coalfields shows a consistent performance drop of 5–12 percentage points in ISP grade accuracy, indicating that region-specific fine-tuning is necessary. This is attributed to rank differences and contrasting depositional environments between Gondwana coalfields.
- The seismic integration approach is limited by the low vertical resolution of seismic data (typically 5–10 m at shallow depths), which may not resolve individual coal seams in multi-seam sequences.

Chapter 5

Conclusion

This study presents a structured AI-driven framework for coal quality estimation using geophysical wireline log data, addressing a long-standing limitation in Indian coal exploration practice. The following principal conclusions are drawn:

1. The hybrid CNN-LSTM model achieves the highest predictive accuracy for GCV estimation (RMSE = 148 kcal/kg, $R^2 = 0.93$, ISP grade accuracy = 83.6%), demonstrating the value of architectures that capture both local depth patterns and sequential stratigraphic context.
2. Bulk density and resistivity logs are the most informative individual predictors for coal quality parameters, supporting the continued prioritization of these log types in coal exploration wireline programmes.
3. Integration of well log-derived quality models with 3D seismic acoustic impedance volumes through co-kriging substantially improves lateral quality prediction, reducing kriging variance by 28% compared to borehole-only interpolation.
4. The framework provides an end-to-end reproducible Python workflow that can be adapted across Indian coalfields, supporting more efficient, cost-effective, and timely coal quality characterization.
5. Critical limitations — including the chemical nature of GCV, thin-seam log resolution, and cross-coalfield transferability — are documented to guide operational implementation decisions.

Future work should focus on incorporating nuclear magnetic resonance (NMR) logs, which are more directly sensitive to hydrogen-bound moisture, and on developing robust domain adaptation strategies for cross-coalfield model transfer. The integration of distributed acoustic sensing (DAS) data and machine learning-based seismic inversion represents a further avenue for improving the continuity and resolution of coal quality models.

The Bhoomathon 2026 framework demonstrates how interdisciplinary collaboration between geologists, geophysicists, and data scientists, facilitated by platforms such as the IAMG-IITB Student Chapter, can accelerate methodological innovation in applied geoscience with direct societal relevance to India's energy security.

References

- [1] Ministry of Coal, Government of India. (2023). Annual Report 2022-23. New Delhi: Ministry of Coal.
- [2] Bureau of Indian Standards. (2022). Indian Standard Practice (ISP) Norms 2022 for Grading of Non-Coking Coal. BIS, New Delhi.
- [3] Chatterjee, R., & Pal, P. K. (2010). Estimation of stress and its use in implementation of proactive roof support system in underground coal mines. *Computers & Geosciences*, 36(9), 1150–1162.
- [4] Mazumdar, B. K. (1954). Theoretical oxygen requirement for coal combustion: Relationship with its calorific value. *Fuel*, 33, 264–266.
- [5] Ellis, D. V., & Singer, J. M. (2007). *Well Logging for Earth Scientists* (2nd ed.). Springer.
- [6] Chopra, S., & Marfurt, K. J. (2007). *Seismic Attributes for Prospect Identification and Reservoir Characterization*. SEG Geophysical Developments Series.
- [7] Bhatt, A., & Helle, H. B. (2002). Committee neural networks for porosity and permeability prediction from well logs. *Geophysical Prospecting*, 50(6), 645–660.
- [8] Zhang, D., Chen, Y., & Meng, J. (2018). Synthetic well logs generation via recurrent neural networks. *Petroleum Exploration and Development*, 45(4), 629–639.
- [9] Sachdev, S. C., Kumar, D., & Sharma, P. (2013). Evaluation of empirical formulae for estimation of gross calorific value of Indian coals. *Journal of Geological Society of India*, 82(5), 491–497.
- [10] Schlumberger. (1991). *Log Interpretation Principles/Applications*. Schlumberger Educational Services, Houston.
- [11] Stach, E., et al. (1982). *Stach's Textbook of Coal Petrology* (3rd ed.). Gebrüder Borntraeger, Berlin.
- [12] Medhurst, T. P., & Brown, E. T. (1998). A study of the mechanical behaviour of coal for pillar design. *International Journal of Rock Mechanics and Mining Sciences*, 35(8), 1087–1105.
- [13] Lomas, H., et al. (2019). Predicting coal quality from geophysical well logs in the Bowen Basin, Australia. *International Journal of Coal Geology*, 203, 1–18.
- [14] Hall, B. (2016). Predicting lithology from well logs with scikit-learn. *The Leading Edge*, 35(6), 500–506.
- [15] Ao, Y., et al. (2019). The linear random forest algorithm and its advantages in machine learning assisted logging interpretation. *Journal of Petroleum Science and Engineering*, 174, 776–789.
- [16] Qian, K., et al. (2020). Application of convolutional neural network in seismic facies identification. *IEEE Transactions on Geoscience and Remote Sensing*, 58(11), 7957–7970.
- [17] Wang, H., et al. (2020). LSTM-based deep learning model for predicting individual mobility traces. *IEEE Transactions on Intelligent Transportation Systems*, 22(6), 3285–3296.
- [18] Ampilov, Y. P., et al. (2009). Seismic inversion for coal quality characterization: Case studies from Western Siberia. *First Break*, 27(7), 47–54.
- [19] Sen, M. K. (2006). *Seismic Inversion*. Society of Petroleum Engineers, Richardson, TX.
- [20] Di, H., et al. (2018). Developing a seismic pattern interpretation network (SpiNet) for automated seismic interpretation. *SEG Technical Program Expanded Abstracts*, 2253–2257.